25901109 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBTG200, MG4DSCBBT201 - MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence

Answer any **10** questions

Each question carries **2** marks

1. Name the purine bases present in DNA. [K] / [CO1]
2. Who provided definitive evidence that DNA is the genetic material of bacteriophages? [K] / [CO1]
3. Identify the non coding region of a gene. [U] / [CO2]
4. Name any 3 arms of tRNA. [K] / [CO3]
5. Name the protein that initiates DNA replication. [K] / [CO3]
6. Which radiation commonly causes thymine dimers? [K] / [CO4]
7. What is the function of the poly-A tail? [K] / [CO3]
8. What enzyme degrades RNA in RNA-DNA hybrid? [K] / [CO3]
9. Who proposed Wobble Hypothesis? [K] / [CO3]
10. Which RNA is modified? [K] / [CO3]

11. Recognize the *lac* operon as an inducible or repressible operon. [K] / [CO4]

12. Name the DNA sequence to which a repressor binds. [K] / [CO4]

[2x10 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **4** marks

13. Explain X-Ray Crystallographic Studies conducted by Rosalind Franklin . [K] / [CO1]

14. Explain the role of hydrogen bonding and base stacking in DNA stability. [U] / [CO1]

15. Explain the structure of a nucleosome. [U] / [CO2]

16. Explain why DNA polymerase requires a primer. [U] / [CO3]

17. Explain the role of the Shine–Dalgarno sequence in initiation. [U] / [CO3]

18. Explain the role of eukaryotic initiation factors (eIFs) in translation. [U] / [CO3]

19. Explain copy-paste transposons. [K] / [CO4]

[4x5 = 20]

Part C

Essay type questions

Answer any **1** questions

Each question carries **10** marks

20. Explain the mechanism of DNA replication in Prokaryotes with a neat labelled diagram. [U] / [CO3]

21. Explain how attenuation controls *trp* operon. [U] / [CO4]

[10x1 = 10]